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SAI VIDYA INSTITUTE OF TECHNOLOGY

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PRINCIPLES OF COMMUNICATION SYSTEMS (15EC45)

IV SEMESTER ECE

MODULE 3 SYLLABUS

RANDOM VARIABLES & PROCESS: Introduction, Probability, Conditional Probability, Random variables, Several Random Variables. Statistical Averages: Function of a random variable, Moments, Random Processes, Mean, Correlation and Covariance function: Properties of autocorrelation function, Cross-correlation functions

NOISE: Shot Noise, Thermal noise, White Noise, Noise Equivalent Bandwidth, Noise Figure

Module-3

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3.1: Random Variables and process :-

Introduction: Random $\rightarrow$ unpredictable.

Randomness, is the essence of communication. In communication system, the signal at any stage belongs to either Deterministic

① Random signals.

The signals are deterministic, if their behaviour is known and can be represented by formula & graph.

The signals are said to be Random (or stochastic) if their behaviour is not known/unpredictable.

This module, gives the overview of probability theory, random variables and various statistical parameters such as mean, variance, moments, standard deviation, autocorrelation, Auto-Variance and cross correlation functions.

Basics of probability:-

- Random Experiment:- Experiments whose outcomes are not predictable are known as random experiments.

Example: $\rightarrow$ Tossing a coin

$\rightarrow$ Throwing a die

$\rightarrow$ Measuring the Noise voltage at the terminals of Resistor.

- Sample Space (S):-

The sample space of a random experiment is a mathematical abstraction used to represent all possible outcomes of the experiment. It is denoted by 'S'.

- Sample point: (s_k)

It represents each outcome of the random experiment. I.e., sample space 'S', consists of any number of sample points & each point represents an outcome of the random experiment.

i.e., $S = \{S_1, S_2, S_3, \dots, S_n\}$ $S_1, S_2, \dots, S_n \Rightarrow$ Sample points in Sample space 'S'.

• Random Event :-

(2)

A Random event is an outcome or Set of outcomes or Set of Sample points that share a common attribute.

Events are denoted by upper case letters such as A, B, etc.

• Mutually exclusive events :- or (Independent events)

Two events A and B are said to be mutually exclusive if they are completely independent or cannot occur together.

• Occurrence of an Event :- An event 'A' of an random experiment is said to have occurred if the experiment gives an outcome that belongs to event 'A'.

• Complement of an event :- The complement of an event A, is the event containing all sample points in "S" but not in A.

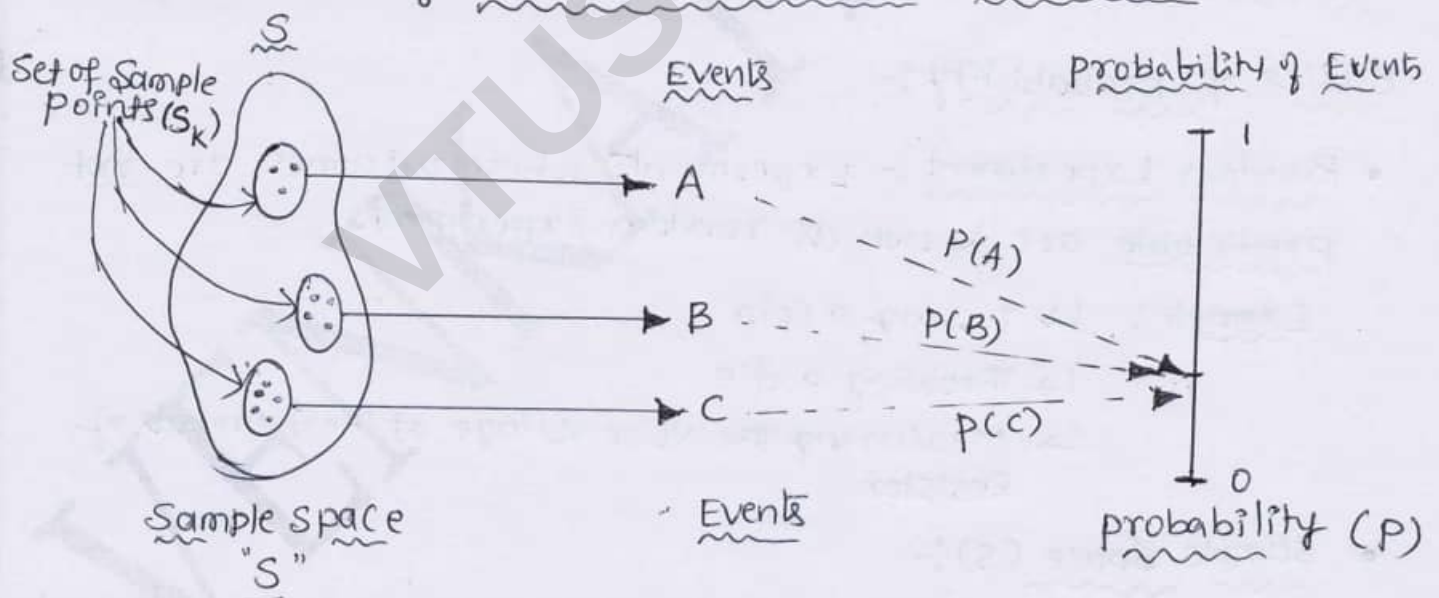


Fig.1: Illustration of Relationship between Sample Space (S), Events and Probability

Figure 1, illustrates the relationship between Sample Space (S), random events (A, B, C) and their probabilities [$P(A)$, $P(B)$, $P(C)$].

• Dependent Events :- Events A, B are said to be dependent if they share common sample points in "S".

3.1: Probability :-

probability of event A, is denoted by $P(A)$.

Let a random experiment is repeated n -times. If the event 'A' occurs n_A -times, then probability of event A, i.e., $P(A)$ is defined as,

$$P(A) = \lim_{n \rightarrow \infty} \left(\frac{n_A}{n} \right)$$

- The ratio, $\frac{n_A}{n}$ represents the fraction of occurrence of event 'A'.

Axioms (Laws) of probability :-

1. $P(A) \geq 0$
2. $P(S) = 1$ [Total probability of all events in Sample Space is 1]
3. If A and B are mutually exclusive events then,

$$P(A+B) = P(A) + P(B)$$

4. If A and B are dependent events then,

$$P(A+B) = P(A) + P(B) - P(AB)$$

where $P(AB) \Rightarrow$ Joint probability of events A, B.

- Joint probability :- [$P(A, B) @ P(AB)$]

Joint probability of two events A and B, is defined as

$$P(A, B) = P(AB) = \lim_{n \rightarrow \infty} \left(\frac{n_{AB}}{n} \right)$$

$\hookrightarrow$ The ratio $\frac{n_{AB}}{n}$, represents the fraction of occurrence of both the events A, B.

3.2: Conditional probability:-

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Let us consider two-events X and Y of a Random Experiment.

- The conditional probability, $P[Y/X]$ refers to the probability of Y -occurring, given that X -has occurred. It is defined as

$$P[Y/X] = \frac{P[X, Y]}{P[X]} \quad \text{--- (1)}$$

where $P[X, Y]$ = Joint probability of events X, Y .

$P[X]$ = probability of event X . [known event]

i.e., Conditional probability is defined as the "ratio of" Joint-probability to that of probability of known event."

Similarly, the Conditional probability, $P[X/Y]$ refers to the probability of X -occurring, given that Y -has occurred. It is defined as,

$$P[X/Y] = \frac{P[X, Y]}{P[Y]} \quad \text{--- (2)}$$

Baye's theorem for Conditional probability:-

It gives the relationship between conditional probabilities $\{P[Y/X], P[X/Y]\}$ and probabilities of individual events $P(A)$ & $P(B)$.

Statement: Bayes theorem states that, the conditional probability,

$$P[Y/X] = \frac{P[X/Y] \cdot P[Y]}{P[X]}$$

proof:- We know that the Conditional probability,

$$P[Y/X] = \frac{P[X, Y]}{P[Y]} \quad \text{--- (1)}$$

$$\therefore P[X, Y] = P[Y/X] \cdot P[X] \text{ --- (2)}$$

$$\text{Similarly } P[X/Y] = \frac{P[X, Y]}{P[Y]} \text{ --- (3)}$$

$$\therefore P[X, Y] = P[X/Y] \cdot P[Y] \text{ --- (4)}$$

By Comparing equations (3) and (4) we get

$$P[Y/X] \cdot P[X] = P[X/Y] \cdot P[Y]$$

$$\therefore P[Y/X] = \frac{P[X/Y] \cdot P[Y]}{P[X]} \leftarrow \text{Bayes theorem}$$

Note: • If X and Y are independent (mutually exclusive) events
 then $\hookrightarrow P[Y/X] = P[Y]$ $\leftarrow$ ^{occurrence of} Event ' Y ' is independent of Event X

$$\hookrightarrow P[X/Y] = P[X]$$

- Therefore Conditional probability is applicable for dependent variables/events.

- For Independent events A, B , Joint probability is

$$P(A, B) = P(AB) = P(A) \cdot P(B)$$

$$P(X, Y) = P(XY) = P(A) \cdot P(B)$$

3.3. Random Variables:-

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Definition:- Random variable is a real valued function, which can take any value from the sample space and its range is set of real numbers.

Random variable may be classified as

- (a) Discrete Random Variable (DRV)
- (b) Continuous Random Variable (CRV)

• Probability mass function ($P[X \leq x]$)

Let X , be a discrete random variable and $x_1, x_2, x_3, \dots, x_n$ be the values of ' X ' can take, then $P[X \leq x]$ is called probability mass function.

- Probability Distribution function:- For any random variable ' X ' probability distribution function is denoted by $F_X(x)$. It is related to probability mass function as

$$F_X(x) = P[X \leq x] \quad \begin{array}{l} X \Rightarrow \text{Random Variable} \\ x = \{x_1, x_2, x_3, \dots, x_n\} \end{array}$$

- The $F_X(x)$ is a function of ' x ' and not a function of random variable ' X '.

↳ properties of distribution function

Distribution function, $F_X(x)$ satisfies the following properties

i) $F_X(x) \geq 0 \quad \because -\infty \leq x \leq \infty$ (Non-Negative)

** ii) $F_X(-\infty) = 0$ and $F_X(\infty) = 1$ [Bounded between 0 & 1]

iii) $P[X \leq a] = F_X(a) \quad \because a \Rightarrow \text{Any real number}$

iv) $P[b < X \leq a] = F_X(a) - F_X(b) \quad \because a > b$

Note: $F_X(x)$ is also known as cumulative distribution function.

** probability density function :-

Q) What do you mean by probability density function? Prove that the total area under the surface of a probability density function (pdf) is always 1.

V.T.U June/July-2017
(5 Marks)

→ Definition: The Derivative of probability distribution function is called as probability density function (pdf). It is denoted by, $f_X(x)$.

Mathematically : $f_X(x) = \frac{d F_X(x)}{dx}$

Where, $F_X(x) \Rightarrow$ probability Distribution function.

$x \Rightarrow$ Random Variable that can take any real value 'x'.

* The total area under the probability density curve is always equal to 1 (unity).

i.e., $\int_{-\infty}^{\infty} f_X(x) dx = 1$

Proof :- By definition,

$$f_X(x) = \frac{d F_X(x)}{dx} \quad \text{--- (1)}$$

Apply Integral on both sides between the limits $-\infty$ to ∞ .

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} \frac{d F_X(x)}{dx} dx$$

$$\int_{-\infty}^{\infty} f_X(x) dx = \left[F_X(x) \right]_{x=-\infty}^{\infty} = F_X(+\infty) - F_X(-\infty)$$

$$\therefore \int_{-\infty}^{\infty} f_X(x) dx = F_X(\infty) - F_X(-\infty) = 1 - 0 = 1$$

(because, $F_X(\infty) = 1$ and $F_X(-\infty) = 0$ (properties of distribution function))

3.4 Several Random Variables :- [More than 1 Random Variable]

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consider two random Variables X and Y .

• Joint distribution function :-

↳ Joint distribution function for two random Variables X, Y is denoted by $F_{X,Y}(x,y)$.

↳ It is defined as the probability that the random Variable X is less than @ equal to ' x ' and that the random Variable Y is less than @ equal to ' y '.

$$\text{i.e., } F_{X,Y}(x,y) = P[X \leq x ; Y \leq y]$$

• Joint probability density function :-

Q) Explain the term joint probability density function of random Variables X and Y . Show that the ~~area~~^{Volume} under the surface of pdf curve is unity?

↳ Joint probability density function of two random Variables X and Y is given by the partial derivative of the joint distribution function. It is denoted by $f_{X,Y}(x,y)$.

$$\text{i.e., } f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \cdot \partial y}$$

where $F_{X,Y}(x,y)$ = Joint probability distribution function of random Variables X, Y .

• Total Volume under the surface of a probability density function is always 1 (unity).

$$\text{i.e., } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx \cdot dy = 1.$$

Proof :- By, definition of Joint probability density function, ⁹

$$f_{x,y}(x,y) = \frac{\partial^2 F_{x,y}(x,y)}{\partial x \cdot \partial y} \quad \text{--- (1)}$$

Apply double integral on both sides between limits $-\infty$ to ∞ .

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{x,y}(x,y) dx \cdot dy &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{\partial^2 F_{x,y}(x,y)}{\partial x \cdot \partial y} \cdot dx \cdot dy \\ &= \left[F_{x,y}(x,y) \right]_{x=y=-\infty}^{x=y=\infty} \\ &= \left[F_{x,y}(\infty, \infty) - F_{x,y}(-\infty, -\infty) \right] \quad \text{--- (2)} \end{aligned}$$

W.K.T using the properties of distribution function,

$$F_{x,y}(\infty, \infty) = 1 \quad \text{and}$$

$$F_{x,y}(-\infty, \infty) = 0$$

$$\begin{aligned} \therefore \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{x,y}(x,y) dx \cdot dy &= \cancel{F_{x,y}(\infty, \infty)}^1 - \cancel{F_{x,y}(-\infty, -\infty)}^0 \\ &= 1 - 0 \\ &= 1 \end{aligned}$$

$\therefore$ Total volume under the surface of a ^{Joint} probability density function is equal to 1. (unity).

Additional problems on Density function & Distribution function 10

Function:-

Formulae :- Distribution function: $F_X(x) = P[X \leq x]$

Density function: $f_X(x) = \frac{d}{dx} F_X(x)$

Properties of distribution function:-

(i) $F_X(\infty) = 1$

(ii) $F_X(-\infty) = 0$

(iii) $P[X \leq a] = F_X(a)$

(iv) $P[b < X \leq a] = F_X(a) - F_X(b) \quad \text{if } a > b$

<1> A Random Variable X has distribution function

$$F_X(x) = \begin{cases} 0 & x < 0 \\ Kx^2 & 0 \leq x \leq 10 \\ 100K & x > 10 \end{cases}$$

(a) Determine the Constant K

(b) Evaluate $P(X \leq a)$ and $P(5 < X \leq 7)$

(c) Determine probability density function.

→ From the given data, probability distribution function

$$F_X(x) = \begin{cases} 0 & x < 0 \\ Kx^2 & 0 \leq x \leq 10 \\ 100K & x > 10 \end{cases}$$

(a) To find Constant 'K' :-

We know that $F_X(\infty) = 1$

from given data $F_X(\infty) = 100K$

$$\therefore 100K = 1 \Rightarrow K = \frac{1}{100}$$

(b) • $P[X \leq 5]$:-

W.K.T $P[X \leq a] = F_X(a)$

$\therefore P[X \leq 5] = F_X(5)$

from given data $F_X(5) = Kx^2 = \left(\frac{1}{100}\right)(5)^2 = \frac{25}{100} = 0.25$
 $\uparrow$ $\uparrow$
 $x=5$ $K\text{-value}$

$\therefore P[X \leq 5] = F_X(5) = 0.25$

• $P[5 < X \leq 7]$:-

W.K.T $P[b < X \leq a] = F_X(a) - F_X(b)$

$\therefore P[5 < X \leq 7] = F_X(7) - F_X(5)$

$F_X(7) = Kx^2 = \frac{1}{100} \times 49 = \frac{49}{100}$ • $F_X(5) = 0.25$

$\therefore P[5 < X \leq 7] = \frac{49}{100} - 0.25 = \underline{\underline{0.24}}$

(c) probability density function :- $[f_X(x)]$

W.K.T $f_X(x) = \frac{dF_X(x)}{dx}$

$$F_X(x) = \begin{cases} 0 & ; x < 0 \\ \frac{x^2}{100} & ; 0 < x \leq 10 \\ 1 & ; x > 10 \end{cases}$$

($\because K = \frac{1}{100}$)

($\because \frac{d}{dx} \left(\frac{x^2}{100} \right) = \frac{2x}{100} = \frac{x}{50}$)

$\therefore f_X(x) = \frac{dF_X(x)}{dx} = \begin{cases} 0 & ; x < 0 \\ \frac{x}{50} & ; 0 < x \leq 10 \\ 0 & ; x > 10 \end{cases}$

3.5. Statistical Averages:-

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Q) With an example, explain what is meant by statistical averages.

V.T.U. June/July 2017

(6-Marks)

→ Statistical average (or) Mean of any random variable is expressed by the summation of the values of random variable 'x' weighted by their probabilities.

Statistical average is also called mean (or) Expectation Value of random variable. It is denoted by $E[X]$ (or) m_x .

$$\text{i.e., } E[X] = m_x = \sum_{\text{all } x} x P[X=x] = \sum_{\text{All } x} x \left(\frac{n_x}{N} \right)$$

Example: Let 'x' be the random variable that is defined on the die experiment, which can take $x = \{1, 2, 3, 4, 5, 6\}$ real values.

Suppose that the die is rolled N-times and the number x appears n_x -times. Then, the statistical average is calculated by summing all the numbers and dividing by the total number of times that the die is rolled.

i.e., Statistical average,

$$E[X] = \sum_{x=1}^6 x \cdot \frac{n_x}{N} \quad \left(\because P[X=x] = \frac{n_x}{N} \right)$$

$$E[X] = \frac{1 \cdot n_1 + 2n_2 + 3n_3 + 4n_4 + 5n_5 + 6n_6}{N}$$

$$\text{i.e., } E[X] = \frac{n_1 + 2n_2 + 3n_3 + 4n_4 + 5n_5 + 6n_6}{N}$$

∴ for Continuous random variable, x statistical average is also calculated by,

$$E[X] = \int x f_x(x) dx$$

Note: Statistical averages includes Mean, Median and Mode.

3.6 : Moments :-

Moments are an application of Random Variable 'X'. There are two types of moments

(a) Moments about the origin

(b) Central moments

(a) Moments about the Origin :-

Let 'X' be a Continuous random Variable, then n^{th} -order moment of random variable 'X' is defined as,

$$E[X^n] = \int_{-\infty}^{\infty} x^n \cdot f_X(x) dx$$

Most widely used moments are,

↳ First moment : for $n=1$; $E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx \leftarrow \begin{matrix} \text{Results} \\ \text{in} \\ \text{Mean} \end{matrix}$

↳ Second moment: for $n=2$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot f_X(x) dx \leftarrow \begin{matrix} \text{Results in Mean Square} \\ \text{Value of Random Variable 'X'} \end{matrix}$$

(b) Central moments :-

moments about the mean value of random variable X are called "Central moments".

n^{th} -order Central moment is defined as

$$E[(X - m_X)^n] = \int_{-\infty}^{\infty} (x - m_X)^n f_X(x) dx$$

* The most widely used Central moment is "Second order Central moment". i.e., for $n=2$

$$E[(X - m_X)^2] = \int_{-\infty}^{\infty} (x - m_X)^2 \cdot f_X(x) \cdot dx$$

* $E[(X - m_X)^2] \Rightarrow \text{Results in Variance of random Variable 'X'}$

3.7: Mean, Variance and Standard deviation:-

a) Mean: Mean Value is also known as Expected Value of Random Variable 'X'. It is the first order moment about the origin. It is denoted by $E[X]$ (or) m_x .

Where $E \rightarrow$ Expectation operator.

Mathematically, mean of Continuous random Variable 'X' is given by,

$$m_x = E[X] = \int_{-\infty}^{\infty} x f_x(x) dx$$

Note: If Random Variable 'X' represents voltage 'v', then mean $\rightarrow$ represents Average DC-value (V_{dc})

b) Variance:-

The Second order Central moment of random Variable 'X' is known as Variance.

Variance of random Variable is denoted by σ_x^2 .

Mathematically it is given by,

$$\sigma_x^2 = E[(X - m_x)^2]$$

Simplification: W.K.T, $(X - m_x)^2 = X^2 + m_x^2 - 2X \cdot m_x$

Where $m_x = \text{Mean of R.V. 'X' = Constant}$

$$\therefore \sigma_x^2 = E[(X - m_x)^2] = E[X^2 + m_x^2 - 2X m_x]$$

$$\sigma_x^2 = E[X^2] + E[m_x^2] - 2E[X \cdot m_x]$$

Since m_x is Constant $E[m_x^2] = m_x^2$ & $2E[X m_x] = 2m_x E[X]$

$$\therefore \sigma_x^2 = E[X^2] + m_x^2 - 2E[X] \cdot m_x$$

W.K.T $E[X] = m_x \leftarrow \text{mean of random variable 'X'}$,

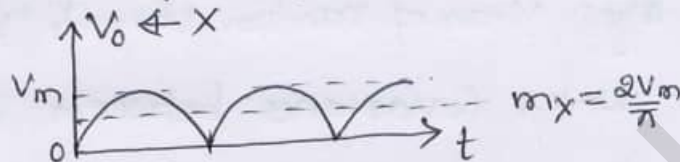
$$\therefore \sigma_x^2 = E[X^2] + m_x^2 - 2m_x \cdot m_x$$

$$\therefore \sigma_x^2 = E[x^2] + m_x^2 - 2m_x^2$$

$$\therefore \sigma_x^2 = E[x^2] - m_x^2$$

i.e., $\text{Var}(x) = \text{Second order moment} - \text{square of Mean about origin}$

Note: If Random Variable 'x' represents output voltage in full-wave rectifier then



Mean $\rightarrow$ Average DC-voltage $= E[x] = m_x$

$E[x^2] \Rightarrow$ Mean square value $\Rightarrow$ It gives total power (Assuming $R=1\Omega$)

$\sigma_x^2 \Rightarrow$ Variance $\Rightarrow$ It gives AC-power.

$m_x^2 \Rightarrow$ square of average DC-voltage $\Rightarrow$ It gives DC-power

$\therefore \sigma_x^2 = E[x^2] - m_x^2$ can be read as

$\Downarrow$
AC-power = Total power - DC-power. (Assuming $R=1\Omega$)

Standard deviation:

square root value of Variance is called as "Standard deviation".

It is denoted by the symbol ' σ_x '.

$$\text{i.e., } \sigma_x = +\sqrt{\text{Var}[x]}$$

Note: If 'x' represents the output voltage of rectifier circuit then standard deviation gives root Mean square Value. (RMS-value).

3.8: Covariance and Correlation Co-efficient :-

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- Covariance :- It gives joint expectation of two-random-variables X and Y . It is denoted by symbol ' λ_{xy} '.

i.e., $\text{cov}(X, Y) = \lambda_{xy} = E[(X - m_x)(Y - m_y)]$

where $m_x = \text{Mean of random Variable 'X'}$
 $m_y = \text{Mean of random Variable 'Y'}$ } Constants.

→ Simplification :- N.K.T Covariance between X, Y is

$$\text{cov}(X, Y) = \lambda_{xy} = E[(X - m_x)(Y - m_y)]$$

$$= E[XY - X m_y - m_x Y + m_x m_y]$$

$$= E[XY] - E[X m_y] - E[m_x Y] + E[m_x m_y]$$

$\uparrow$ Constant $\uparrow$ Constant $\uparrow$ Constant

$$= E[XY] - m_y E[X] - m_x E[Y] + m_x m_y$$

$$= E[XY] - m_y m_x - m_x m_y + m_x m_y$$

$$= E[XY] - 2m_x m_y + m_x m_y$$

$$\boxed{\text{cov}(X, Y) = \lambda_{xy} = E[XY] - m_x m_y} \quad \text{--- (1)}$$

Note :- • If $E[XY] = 0$, then the random Variables X, Y are said to be Orthogonal.

- If two random Variables X and Y are Independent then they are said to be "Uncorrelated".
- For Independent Random Variables, X and Y
 $E[XY] = E[X] \cdot E[Y] = m_x \cdot m_y$ and Covariance is zero (using equation (1))

• Correlation-co-efficient :-

Correlation Co-efficient is a measure of dependancy between random variables X and Y .

It is defined as the ratio of Covariance (λ_{xy} or $\text{Cov}(x,y)$) to the product ($\sigma_x \cdot \sigma_y$).

It is denoted by symbol ρ_{xy} .

i.e., Correlation Co-efficient is given by

$$\rho_{xy} = \frac{\text{Cov}(X,Y)}{\sigma_x \cdot \sigma_y} ; \quad \text{where}$$

σ_x = Standard deviation of X
 σ_y = Standard deviation of Y .

$$\therefore \rho_{xy} = \frac{E[XY] - m_x m_y}{\sigma_x \cdot \sigma_y}$$

It gives dependancy between random variables X and Y .

i.e., (i) If $\rho_{xy} = 0 \Rightarrow$ Random Variables X and Y are said to be Completely Independent

(ii) If $\rho_{xy} = 1 \Rightarrow X$ and Y are strongly dependent on each other.

(iii) If $0 < \rho_{xy} < 1 \Rightarrow X$ and Y are said to neither independent nor totally dependent on each other.

3.9: Auto Correlation function:- (VTU Q.P)

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Q) Define autocorrelation function. Mention its important properties.
V.T.U Dec 2017/Jan. 2018 (6 Marks)

↳ Definition: The auto correlation function of the random variable 'X', is defined as the expectation of the product of two random variables, $X(k)$ and $X(l)$, obtained by observing the random variable 'X' at times ' k ' and ' l ' respectively.

It is denoted by $\gamma_X(k, l)$. @ $\gamma_X(\tau)$

i.e., - Auto Correlation function. is given by

$$\gamma_X(k, l) = E[X(k) \cdot X(l)] = \gamma_X(k-l)$$

∴ Auto correlation function gives correlation of a signal with itself but delayed in time.

- Auto correlation function, $\gamma_X(k, l)$ is a function of time difference $(k-l)$.

Properties:-

Let $\gamma_X(\tau)$ be the auto correlation function of Random process 'X' (Variable)

then

<i> $\gamma_X(\tau) = \gamma_X(k-l)$ ∴ It is a function of time difference $(k-l)$

<ii> $\gamma_X(0) = E[X^2]$ ∴ when $k=l$.

<iii> $\gamma_X(\tau)$ is maximum value at $\tau=0$.

i.e., $\gamma_X(0) \geq \gamma_X(\tau)$ for any value of ' τ '

<iv> $\gamma_X(\tau)$ is a even function of (τ)

i.e., $\gamma_X(\tau) = \gamma_X(-\tau)$

* Auto Covariance :- (VTU Q.P)

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- Auto Covariance function of random process 'X' is defined as,

$$C_X(k, l) = E[(X(k) - m_X) \cdot (X(l) - m_X)]$$

$$C_X(k, l) = \gamma_X(k-l) - m_X^2$$

Where $C_X(k, l)$ = Auto Covariance

$\gamma_X(k-l)$ = Auto Correlation between $X(k)$ and $X(l)$

m_X = Mean of random variable 'X'.

3.10 :- Cross Correlation function :-

(a) Define Cross Correlation function. Mention its properties.

(8 Marks)

VTU Q.P.

↳ Definition:- Consider two random variables X and Y, observed at time instants 'k' and 'l' respectively, then the cross correlation function between random variables X and Y is given by

$$\gamma_{XY}(k, l) = E[X(k) \cdot Y(l)]$$

(b)

$$\gamma_{YX}(l, k) = E[Y(l) \cdot X(k)]$$

∴ cross correlation function gives the correlation between two different random processes.

↳ Properties:-

1. It is symmetric function.

$$\text{i.e., } \gamma_{XY}(\tau) = \gamma_{YX}(-\tau)$$

2. It does not have maximum at origin

$$3. |\gamma_{XY}(\tau)| \leq \frac{1}{2} [\gamma_X(0) + \gamma_Y(0)]$$

where $\gamma_X(0)$ & $\gamma_Y(0)$ are auto-correlation functions of X and Y respectively at $\tau=0$

Numerical problem on Mean, Variance & Standard deviation :-

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Formulae: • $m_x = E[x] = \int_{-\infty}^{\infty} x f_x(x) dx \Rightarrow \text{Mean}$

• $\text{Var}(x) = E[x^2] - m_x^2 \Rightarrow \text{Variance } (\sigma_x^2)$

• Standard deviation: $\sigma_x = +\sqrt{\text{Var}(x)}$

① A random variable 'x', has distribution function

$$F_x(x) = \begin{cases} 0 & ; x < 0 \\ \frac{x}{8} & ; 0 \leq x \leq 2 \\ \frac{x^2}{16} & ; 2 \leq x \leq 4 \\ 1 & ; 4 \leq x \end{cases}$$

Determine (a) Mean ($\bar{x}$) (b) Variance (c) Standard deviation.

→ Ans:- from given data, probability distribution function,

$$F_x(x) = \begin{cases} 0 & ; x < 0 \\ x/8 & ; 0 \leq x \leq 2 \\ x^2/16 & ; 2 \leq x \leq 4 \\ 1 & ; x \geq 4 \end{cases}$$

∴ probability density function $f_x(x)$ is,

$$f_x(x) = \frac{dF_x(x)}{dx} = \begin{cases} 0 & ; x < 0 \\ 1/8 & ; 0 \leq x \leq 2 \\ x/8 & ; 2 \leq x \leq 4 \\ 0 & ; x \geq 4 \end{cases}$$

We can find Mean, Variance & Standard deviation using density function $f_x(x)$.

(a) Mean: $E[x] = \int_{-\infty}^{\infty} x f_x(x) dx = \int_0^2 x \cdot \left(\frac{1}{8}\right) dx + \int_2^4 x \cdot \left(\frac{x}{8}\right) dx$
 $\therefore E[x] = \underline{2.583}$

(b) Variance: N.K.T $\text{Var}(x) = E[x^2] - m_x^2$

$$\therefore E[x^2] = \int_{-\infty}^{\infty} x^2 f_x(x) dx = \int_0^2 x^2 \cdot \left(\frac{1}{8}\right) dx + \int_2^4 x^2 \cdot \left(\frac{x}{8}\right) dx = \underline{7.833}$$

$$\therefore \text{Var}(x) = \sigma_x^2 = E[x^2] - m_x^2 = 7.833 - (2.583)^2 = \underline{1.161}$$

(c) Standard deviation: $\sigma_x = +\sqrt{\text{Var}(x)} = \underline{1.077}$

Module 3: (b): Noise :-

Q) Define Noise. Mention its various types. (VTU Q.P 6 Marks)

→ Noise is a disturbance @ unwanted frequency signal which appears within the required frequency signal.

The classification of Noise-sources are shown in Fig.1.

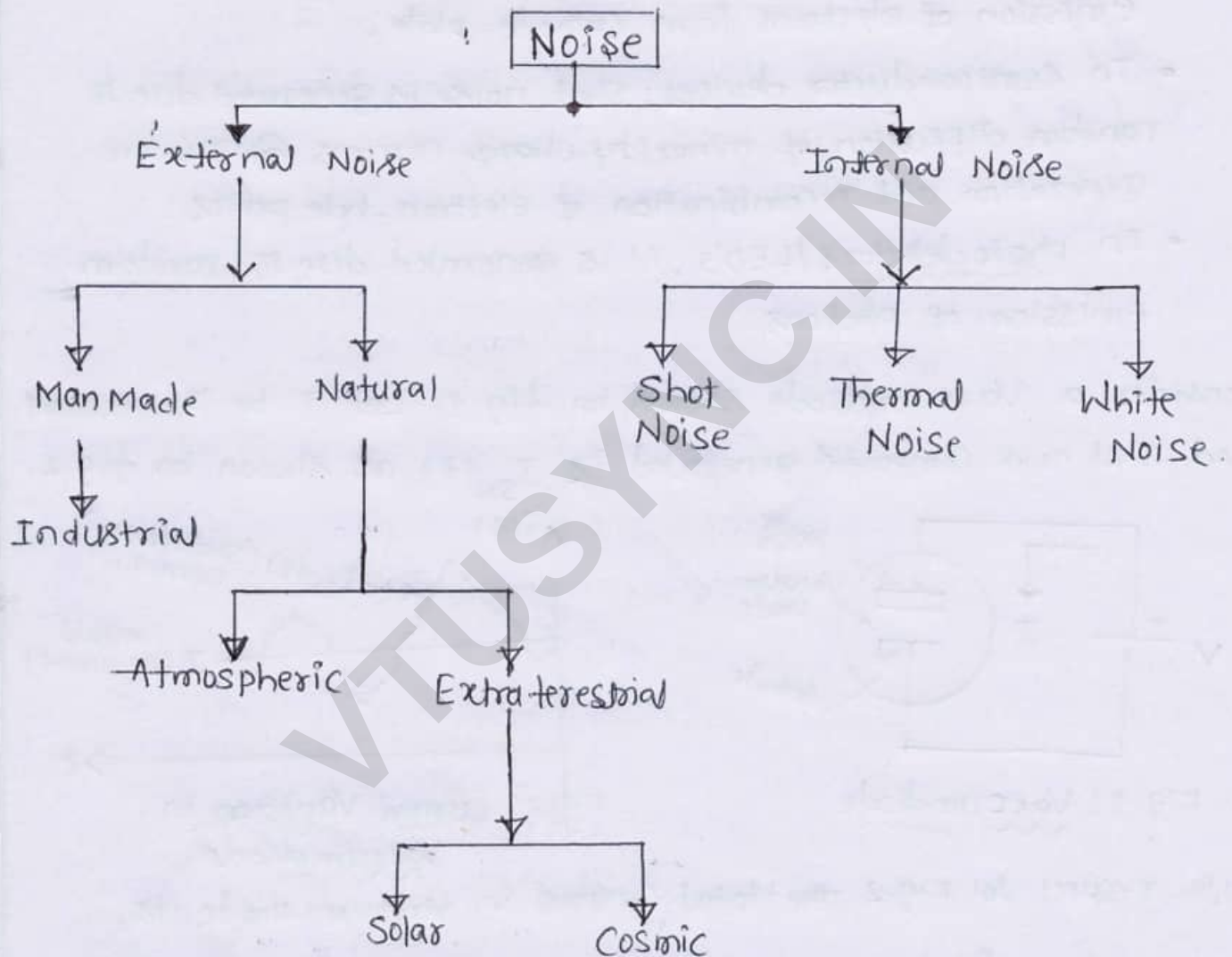


Fig1: Classification of Noise-sources/signals

Shot Noise: Shot Noise appears in active devices due to random behaviour of charge carriers (electrons and holes).

Thermal Noise: Thermal Noise is generated to random movement of thermally electrons.

White Noise: The Noise which is gaussian distribution and has flat @ constant power spectral density over wide range of frequencies.

*** V.T.U.Q.P

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3.11: Shot Noise :-

(Q) Write a short note on shot noise.

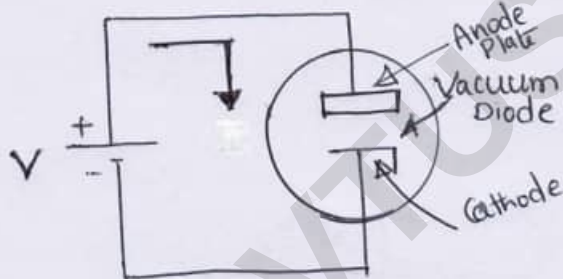
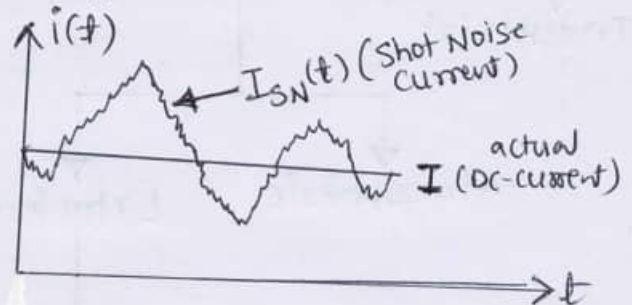
4-Marks.

→ Shot noise appears in active devices due to random behaviour of charge carriers (electrons and holes).

Example:

- In Vacuum tubes, shot noise is generated due to random emission of electrons from Cathode-plate.
- In semiconductor devices, shot noise is generated due to random diffusion of minority charge carriers @ random generation and recombination of electron-hole pairs.
- In photodetectors / LED's, it is generated due to random emission of photons.

Consider a Vacuum diode shown in fig.1. Let 'I' be the current and shot noise component generated is $I_{SN}(t)$ as shown in fig.2.

Fig.1: Vacuum diodeFig.2: current Variation in Vacuum diode

With respect to Fig.2. the total current in vacuum diode is,

$$i(t) = I + I_{SN}(t) \quad \therefore I_{SN}(t) \Rightarrow \text{shot noise current generated due to random emission of electrons from Cathode.}$$

∴ The Mean Square Value of Fluctuating, shot noise current in Vacuum diode is given by

$$E[I_{SN}^2] = 2qIB_N \quad \text{Where } q = 1.6 \times 10^{-19} \text{ Coulombs}$$

$B_N = \text{Noise equivalent Bandwidth}$

→ for pN-junction diodes,

$$E[I_{SN}^2] = 2q(I + I_S)B_N$$

$I = \text{required DC-current in Vacuum diode.}$

$I_S = \text{Reverse Leakage current} \approx 0$

3.12: Thermal Noise:- (VTU Q.P)

a) Write a short note on Thermal Noise.

↳ Thermal Noise is generated due to random motion of thermally induced carriers (electrons) in a conductor.

↳ The random motion of thermally induced electrons produces electric current which is random in nature. This random current is called "thermal noise". @ "Johnson Noise"

↳ Figure 1 shows noise-model using resistor, and its equivalent thevenin's circuit in fig. 2.

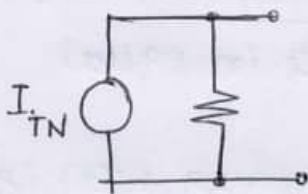


Fig 1: Noise model

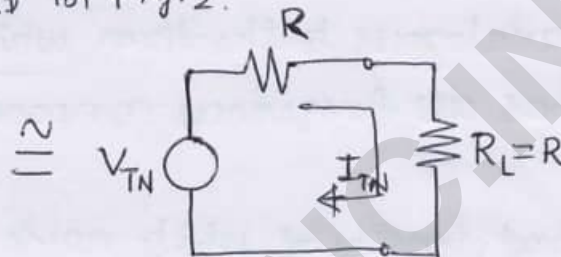


Fig 2: Equivalent circuit (To find Max Noise Power)
 $R_L = R$

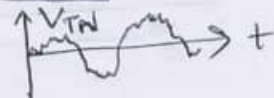
I_{TN} = Thermal Noise Current

$V_{TN} = I_{TN} \times R$
↳ Thermal Noise Voltage

↳ The Mean Value of thermal noise current is always zero.

↳ Mean square value of the thermal noise voltage is

$$E[V_{TN}^2] = 4 K B_N T R$$



where K = Boltzmann Constant = 1.38×10^{-28} J/s

T = temperature at which resistor is operating = 290 K (standard)

B_N = Noise equivalent Bandwidth in Hz.

R = Resistor in Ω

↳ The Maximum noise power, produced across noisy resistor model shown in fig. 2 is

$$P_N = \frac{E[V_{TN}^2]}{4R} = \frac{4 K B_N T R}{4R} = \underline{\underline{K T B_N}} \text{ Watts.}$$

Note: Max. power delivered to load when $R_L = R$ &

$$P_{max} = \frac{V^2}{4R} //$$

3.13 : White Noise:- (VTU Q.P)

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Q) Define white noise. plot power spectral density and auto correlation function (ACF) of white noise

VTU June/July 2017 - 5 Marks -

→ White Noise:-

The Noise which has Gaussian distribution and have flat power spectral density over a wide range of frequencies is called white noise.

- It is denoted by $w(t)$.
- White Noise is analogous to the term "white light" in the sense that it has all frequency components in equal amounts.

→ The power spectral density of white noise process, $w(t)$ is

$$S_w(f) = \frac{N_0}{2} \quad \text{--- (1)}$$

where

$$N_0 = k T_e$$

T_e = Noise equivalent temperature

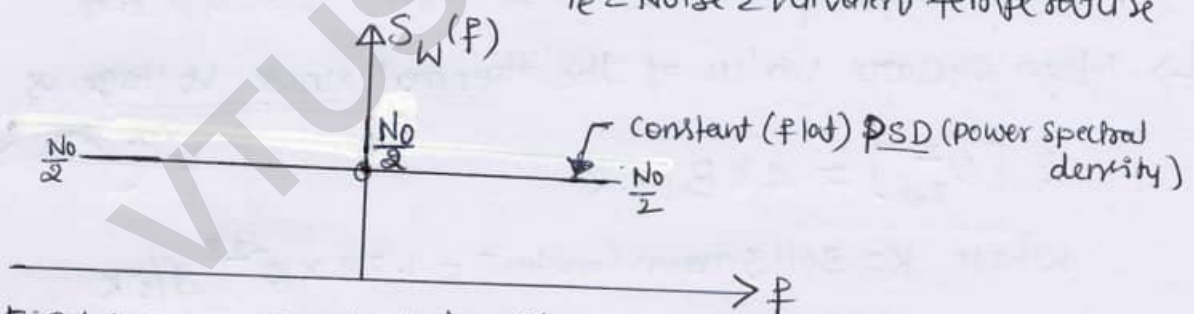


Fig 1: power spectral density of white noise:-

→ Auto-correlation function of white noise, $w(t)$ is

$$\gamma_w(\tau) = \frac{N_0}{2} \delta(\tau) \quad (\text{i.e., } S_w(f) \xleftrightarrow{\text{IFT}} \gamma_w(\tau))$$

$$\therefore \gamma_w(\tau) = \begin{cases} \frac{N_0}{2} & \tau = 0 \\ 0 & \text{elsewhere} \end{cases}$$

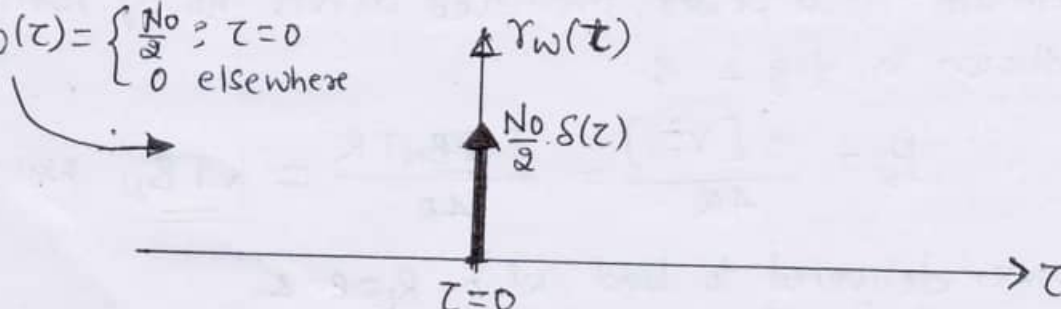


Fig 2: Auto correlation function of white noise

3.14. Noise Equivalent Bandwidth

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Q) What is noise equivalent Bandwidth? Derive an expression for the same.

VTU June/July - 2017 (5 Marks)

↳ Noise equivalent bandwidth represents the frequency selectivity of the filter.

It is denoted by symbol ' B_N ' Hz.

Mathematically, it is defined as the ratio of total noise - output power to that of Noise spectral density at input.

$$\text{i.e., } B_N = \frac{P_{no}}{S_{ni}} \quad (*)$$

Where, P_{no} = Total noise output power in Watts.

S_{ni} = Input Noise spectral density in Watts/Hz.

Equation (1) is simplified to, $B_N = \frac{1}{4RC}$ Hz.

Proof:- consider a Low pass filter (LPF) having voltage ratio transfer function $|H(j\omega)|$, as shown in Fig. 1 and its equivalent circuit in Fig. 2.



Fig 1: LPF with i/p & o/p spectral density

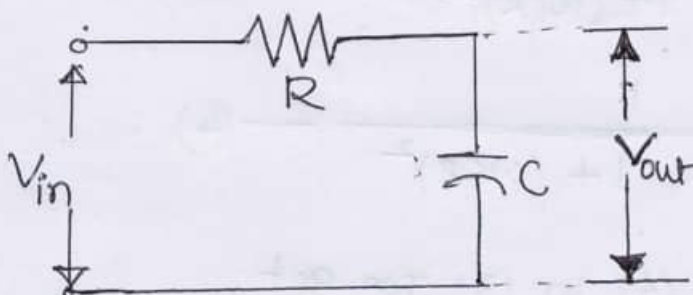


Fig 2: RC-LPF with input and output voltage

We know that Noise bandwidth B_N , is given by

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$$B_N = \frac{P_{n0}}{S_{ni}} \rightarrow (1)$$

Input noise Spectral density, $S_{ni} = KT \rightarrow (2)$

K = Boltzmann's Constant

To find total Noise output power: (P_{n0})

From Fig. 1. the Voltage transfer function, $H(j\omega)$ in terms of input and output noise Spectral density S_{ni} & S_{no} is

$$H(j\omega) = \sqrt{\frac{S_{no}}{S_{ni}}} \Rightarrow |H(j\omega)|^2 = \frac{S_{no}}{S_{ni}}$$

$$\therefore S_{no} = |H(j\omega)|^2 \cdot S_{ni}$$

$$S_{no} = |H(j\omega)|^2 \cdot \downarrow \cdot KT \rightarrow (3)$$

From Fig. 2, RC-LPF the Voltage transfer function,

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{1/sC}{R + 1/sC} = \frac{1/sC}{\frac{SCR + 1}{SC}} = \frac{1}{1 + SCR}$$

$$\therefore H(s) = \frac{1}{1 + SCR} ; s = j\omega$$

$$\therefore H(j\omega) = \frac{1}{1 + j\omega CR}$$

Take magnitude on both sides, we get

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

$$\therefore |H(j\omega)|^2 = \frac{1}{1 + (\omega CR)^2} \rightarrow (4)$$

Using equation (4) in (3) we get,

$$S_{n_0} = \frac{1}{1 + (\omega CR)^2} \cdot KT$$

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$$S_{n_0} = \frac{KT}{1 + (\omega CR)^2} \quad \text{--- (5)}$$

∴ The total noise output power, P_{n_0} is

$$P_{n_0} = \int_0^{\infty} S_{n_0} \cdot df$$

$$P_{n_0} = \int_0^{\infty} \frac{KT}{1 + (\omega CR)^2} \cdot df$$

$$\omega = 2\pi f \quad \omega \cdot K \cdot T$$

$$P_{n_0} = \int_0^{\infty} \frac{KT}{[1 + (2\pi f CR)^2]} \cdot df \quad \text{--- (6)}$$

Let $2\pi f CR = x$ then $(2\pi f CR)^2 = x^2$

differentiate 'f' with respect to 'x' we get

$$2\pi CR \frac{df}{dx} = 1 \Rightarrow df = \frac{dx}{2\pi CR}$$

$$P_{n_0} = \int_0^{\infty} \frac{KT}{(1+x^2)} \cdot \frac{dx}{2\pi CR} = \frac{KT}{2\pi CR} \int_0^{\infty} \frac{1}{1+x^2} \cdot dx$$

$$P_{n_0} = \frac{KT}{2\pi CR} [\tan^{-1}(x)]_0^{\infty} \quad \left(\because \int \frac{1}{1+x^2} = \tan^{-1} x \right)$$

$$P_{n_0} = \frac{KT}{2\pi CR} [\tan^{-1}(\infty) - \tan^{-1}(0)] = \frac{KT}{2\pi CR} \left(\frac{\pi}{2} - 0 \right)$$

$$P_{n_0} = \frac{KT}{2\pi CR} \times \frac{\pi}{2} = \frac{KT}{4CR} \quad \text{--- (7)}$$

Using equations (7) & (2) in equation (1) we get

$$B_N = \frac{P_{n_0}}{S_{n_i}} = \frac{KT}{4CR \times KT} = \frac{1}{4CR} \quad \left(\because S_{n_i} = KT \right)$$

hence the proof.

Studied smart, not hard — thanks to VTUSync.in

3.15 : Noise Factor and Noise Figure:-

Noise factor, of an any amplifier or network is defined as the ratio of signal to noise power ratio at the input and to that of signal to noise power at the output.

i.e., Noise factor, $F = \frac{(S/N) \text{ power ratio at the input}}{(S/N) \text{ power ratio at the output}}$

$$\text{i.e., } F = \frac{(P_{Si} / P_{Ni})}{(P_{So} / P_{No})} = \frac{P_{Si}}{P_{Ni}} \times \frac{P_{No}}{P_{So}} \rightarrow (1)$$

P_{Si} = Signal power at input in Watts.

P_{Ni} = Noise power at input in Watts.

P_{So} = Signal power at output in Watts.

P_{No} = Noise power at output in Watts.

Noise Figure:- (F_{dB})

Noise factor, F expressed in dB, is known as Noise Figure.

i.e., $\text{Noise Figure} = 10 \log [F]$

②

$$\text{Noise Figure} = 10 \log \left[\frac{(S/N)_{\text{input}}}{(S/N)_{\text{output}}} \right]$$

$$F_{dB} = 10 \log (S/N)_{\text{input}} - 10 \log (S/N)_{\text{output}}$$

$\underbrace{\hspace{1.5cm}}_{(S/N)_{\text{input}} \text{ in dB}}, \quad \underbrace{\hspace{1.5cm}}_{(S/N)_{\text{output}} \text{ in dB}}$

Note : The ideal value of noise figure is 0 dB

* Equivalent Noise temperature $\langle T_e \rangle$:-

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The equivalent Noise temperature is given by,

$$T_e = (F-1) T$$

where, F = Noise factor

T = Temperature in kelvin = 290K (standard value) (T_0).

* Cascade Connection of two port networks :- (6R)

<Friis's Formula for Amplifiers Connected in Cascade> ^{VTU Q.P (6 marks)}

Consider cascade connection of Noisy two-port networks (Amplifiers) shown in fig.1.

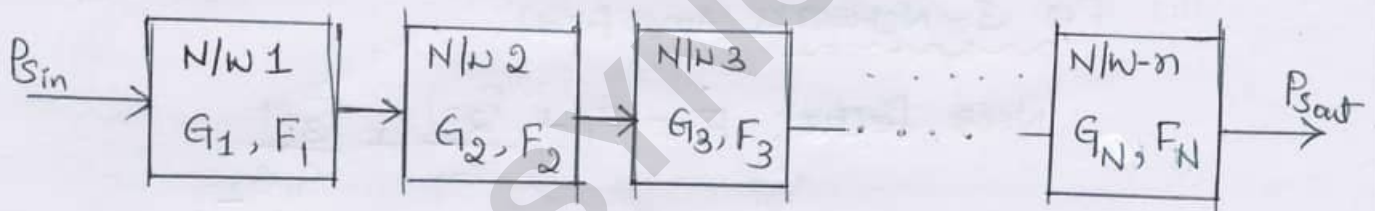


Fig.1: Cascade Connection of N-noise networks (Amplifiers)

Where G_N = Gain of Network-N

F_N = Noise factor of Network-N

The overall noise factor, F for N-amplifiers (networks) connected in cascade is given by Friis-Formula,

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots \rightarrow (1)$$

and its equivalent noise temperature is given by

$$T_e = (F-1) T$$

where 'F' is given by equation (1)

Note: For 2-Amplifiers Connected in Cascade $F = F_1 + \frac{F_2 - 1}{G_1}$

Studied smart, not hard — thanks to VTUSync.in

Numerical problems:-

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List of Formulae:-

1. Noise Figure: $F_{dB} = 10 \log (F)$: $F = \text{Noise factor} = \frac{(S/N)_{\text{input}}}{(S/N)_{\text{output}}}$ where

2. Noise equivalent Temperature : $T_e = (F-1) T$

3. Friis formula for cascaded Networks/Amplifiers:-

(i) For 2- Networks (Amplifiers)

$$\text{Noise factor: } F = F_1 + \frac{F_2 - 1}{G_1}$$

$$\text{Noise Figure: } F_{dB} = 10 \log (F)$$

(ii) For 3- Networks (Amplifiers)

$$\text{Noise factor: } F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2}$$

$$\text{Noise Figure: } F_{dB} = 10 \log (F)$$

4. Noise equivalent Bandwidth:-

$$B_N = \frac{1}{4RC} \text{ Hz.}$$

5. Shot Noise:-

Mean square value of shot noise current is

$$E[I_{SN}^2] = 2q I B_N \quad ; \quad q = 1.6 \times 10^{-19} \text{ Coulombs}$$

6. Thermal Noise:-

• Mean square value of Thermal Noise voltage is

$$E[V_{TN}^2] = 4K B_N T R \quad \text{and} \quad K = \text{Boltzmann's Constant}$$

$$K = 1.38 \times 10^{-28}$$

$$\text{Noise power: } P_N = K T B_N$$

1. Suppose amplifier 1 has a noise figure of 9 dB and power gain of 15 dB. It is connected in cascade to the amplifier 2 with noise figure of 20 dB. Calculate the overall noise figure for this cascade connection in decibel units

VTU, DEC-2017/JAN-2018

→ Given data:-

4-Marks.

$$F_{1\text{ dB}} = 9 \text{ dB} ; G_{1\text{ dB}} = 15 ; F_{2\text{ dB}} = 20 ; F_{\text{dB}} = ?$$

$$\therefore F_1 = 10^{\left(\frac{9}{10}\right)} ; G_1 = 10^{\left(\frac{15}{10}\right)} ; F_2 = 10^{\left(\frac{20}{10}\right)}$$

$$\boxed{F_1 = 7.943} ; \boxed{G_1 = 31.623} ; \boxed{F_2 = 100}$$

Using Friis's formula for 2-amplifiers connected in cascade,

$$\text{overall noise factor, } F = F_1 + \frac{F_2 - 1}{G_1} = 7.943 + \frac{100 - 1}{31.623}$$

$$\therefore F = 11.073$$

$$\therefore \text{overall noise figure, } F_{\text{dB}} = 10 \log(F) = 10 \log(11.073)$$

$$\boxed{F_{\text{dB}} = 10.443 \text{ dB}}$$

2. Two port devices are connected in cascade. For the first stage, the noise figure and available power gain are 5 dB and 12 dB respectively. For the second stage, the noise figure and available power gain are 15 dB and 10 dB respectively. Determine the overall noise figure in dB

June/July-2016

(6 Marks)

→ Given data: $\boxed{N_1/M_1} - \boxed{N_2/M_2}$

$$F_{1\text{ dB}} = 5 \Rightarrow F_1 = 10^{\left(\frac{5}{10}\right)} = 3.162$$

$$G_{1\text{ dB}} = 12 \Rightarrow G_1 = 10^{\left(\frac{12}{10}\right)} = 15.85$$

$$F_{2\text{ dB}} = 15 \Rightarrow F_2 = 10^{\left(\frac{15}{10}\right)} = 31.623$$

$$G_{2\text{ dB}} = 10 \Rightarrow G_2 = 10^{\left(\frac{10}{10}\right)} = 10$$

∴ overall noise factor,

$$F = F_1 + \frac{F_2 - 1}{G_1} = 3.162 + \frac{31.623 - 1}{15.85}$$

$$\therefore \boxed{F = 5.094}$$

∴ overall noise figure,

$$F_{\text{dB}} = 10 \log(F) = 7.07 \text{ dB} //$$

- (3) In a TV-receiver, the antenna is often mounted on a tall mast and a long cable is used to connect the antenna, to the receiver. To overcome the effect of lossy cable, a pre-amplifier is mounted on the antenna as shown in figure 3. (32)

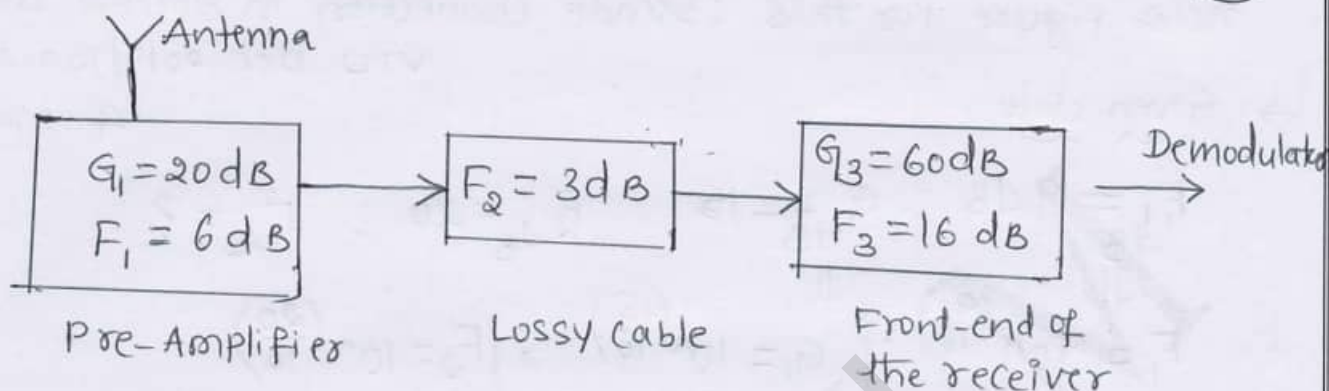


Figure 3: TV-receiver

- (a) Find the overall noise figure of the system
 (b) Find the overall noise figure of the system, if the pre-amplifier is omitted and the gain of front end receiver is increased by 20 dB.

→ Given data :-

(i) pre-Amplifier:

$$G_{1dB} = 20 \Rightarrow G_1 = 10^{\left(\frac{20}{10}\right)} = 100$$

$$F_{1dB} = 6 \Rightarrow F_1 = 10^{\left(\frac{6}{10}\right)} = 3.98$$

(ii) Lossy cable:-

$$F_{2dB} = 3 \Rightarrow F_2 = 10^{\left(\frac{3}{10}\right)} \approx 2$$

$$* G_2 = \frac{1}{F_2} = \frac{1}{2} = 0.5$$

Gain for
Lossy cable

(iii) Front-end of the receiver:-

$$G_{3dB} = 60 \Rightarrow G_3 = 10^{\left(\frac{60}{10}\right)} = 1000$$

$$F_{3dB} = 16 \Rightarrow F_3 = 10^{\left(\frac{16}{10}\right)} = 39.81$$

(a) overall Noise Figure of the system:-

for 3-networks Friis formula is

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2}$$

$$F = 3.98 + \frac{(2-1)}{100} + \frac{(39.81-1)}{100 \times 0.5}$$

$$F = 4.7662$$

$$\therefore F_{dB} = 10 \log(F) = 6.78 \text{ dB}$$

(b) If pre-amplifier is omitted:-

$$F = F_2 + \frac{F_3 - 1}{G_2} \rightarrow \begin{array}{|c|c|} \hline F_2 & F_3 \\ \hline G_2 & G_3 \\ \hline \end{array} \rightarrow$$

$$F = 79.62$$

$$\therefore F_{dB} = 10 \log F = 19.01 \text{ dB}$$

A> The signal power and noise power measured at the input of an amplifier are $150\mu\text{W}$ and $1.5\mu\text{W}$ respectively. If the signal power at the output is $1.5\mu\text{W}$ and noise power is 40mW calculate the amplifier noise factor and noise figure. (33)

↳ Given data:

$$P_{Si} = 150\mu\text{W}$$

$$P_{Ni} = 1.5\mu\text{W}$$

$$P_{So} = 1.5\mu\text{W}$$

$$P_{No} = 40\text{mW}$$

$$F = ?$$

$$F_{dB} = ?$$

W.K.T the Noise factor in terms of signal, noise power ratio is given by

$$F = \frac{P_{Si}}{P_{Ni}} \times \frac{P_{No}}{P_{So}} = \frac{150 \times 10^{-6}}{1.5 \times 10^{-6}} \times \frac{40 \times 10^{-3}}{1.5 \times 10^{-6}}$$

$$\therefore F = 2.666$$

$$\therefore \text{Noise Figure, } F_{dB} = 10 \log(F) = 4.26 \text{ dB}$$

S> The SNR at the input of an amplifier is 40dB . If the noise figure of an amplifier is 20dB , calculate the signal to noise ratio in dB at the amplifier output

↳ Given data:-

$$(SNR)_{\text{input}} = 40 \text{ dB}$$

$$F_{dB} = 20$$

$$(SNR)_{\text{output}} = ?$$

W.K.T

Noise factor, F in terms of (SNR) ratio is

$$F = \frac{(SNR)_{\text{input}}}{(SNR)_{\text{output}}} \Rightarrow 10 \log F = 10 \log \left(\frac{(SNR)_{\text{input}}}{(SNR)_{\text{output}}} \right)$$

$$\therefore F_{dB} = (SNR)_{\text{input}} \text{ dB} - (SNR)_{\text{output}} \text{ dB}$$

$$20 = 40 - (SNR)_{\text{output}} \text{ dB}$$

$$\therefore (SNR)_{\text{output}} = 20 \text{ dB}$$

6> A signal circuit is equivalent to a parallel combination of $R = 1\text{k}\Omega$ and $C = 0.47\text{MF}$. Calculate the effective noise bandwidth.

$$R = 1\text{k}\Omega$$

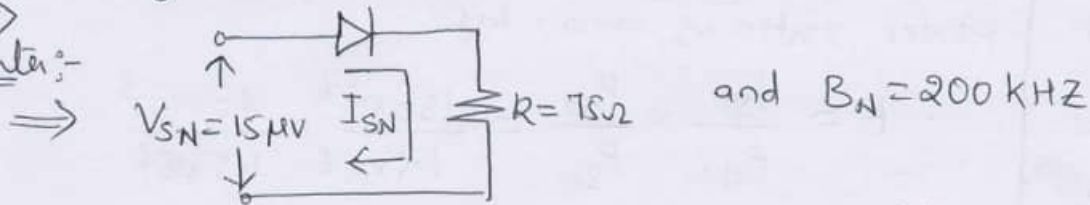
$$C = 0.47\text{MF}$$

W.K.T Noise equivalent bandwidth (effective Noise Bandwidth)

$$B_N = \frac{1}{4RC} = \frac{1}{4 \times 1 \times 10^3 \times 0.47 \times 10^{-6}} = 531.915 \text{ Hz}$$

(T) A noise generator using diode is required to produce $15 \mu\text{V}$ noise voltage in a receiver which has an input impedance of 75Ω (purely resistive). The receiver has a noise power bandwidth of 200 KHz . Calculate the current through the diode.

↳ Data:-



$$\therefore \text{Shot noise current in diode is: } I_{SN} = \frac{V_{SN}}{R} = \frac{15 \times 10^{-6}}{75} = 2 \times 10^{-7} \text{ A}$$

The actual current, I , through the diode is given by

$$E[I_{SN}^2] = 2qIB_N$$

$$\Downarrow$$

$$(2 \times 10^{-7})^2 = 2 \times 1.6 \times 10^{-19} \times I \times 200 \times 10^3$$

$$\therefore I = \frac{(2 \times 10^{-7})^2}{2 \times 1.6 \times 10^{-19} \times 200 \times 10^3} = \underline{\underline{6.25 \text{ mA}}}$$

(S) Calculate the rms noise voltage and thermal noise power appearing across a $20 \text{ k}\Omega$ resistor at 25°C temperature, with an effective noise Bandwidth of 10 KHz .

↳ Given: $R = 20 \text{ k}\Omega$; $T = 25^\circ \text{C} = 273 + 25 = 298 \text{ K}$; $B_N = 10 \text{ KHz}$

(i) rms noise voltage :- $(\sqrt{E[V_{TN}^2]})$ (Root Mean Square Value of V_{TN})

$$\text{N.K.T } E[V_{TN}^2] = 4KT B_N R$$

$$E[V_{TN}^2] = 4 \times 1.38 \times 10^{-23} \times 298 \times 10 \times 10^3 \times 20 \times 10^3$$

$$E[V_{TN}^2] = 3.2899 \times 10^{-12} \text{ Volt}^2$$

$$\therefore \sqrt{E[V_{TN}^2]} = \underline{\underline{1.8138 \mu\text{V}}} \leftarrow \text{RMS noise voltage.}$$

(ii) Thermal noise power :-

$$P_N = KTB_N = 1.38 \times 10^{-23} \times 298 \times 10 \times 10^3$$

$$P_N = \underline{\underline{4.11 \times 10^{-17} \text{ Watts}}}$$